S9 – Sensitivity Analysis – AHP Weight Perturbation (Governance-Gate Stability)

Purpose

The purpose of this analysis is to evaluate whether moderate imprecision in the Analytic Hierarchy Process (AHP)-derived weights materially alters the governance-gate outcomes associated with the calibrated HIMS-CDI model. This supplementary experiment addresses potential concerns regarding the inherent subjectivity of expert-elicited weighting schemes.

Methodology

The analysis commenced using the baseline normalised AHP weights adopted within the governance taxonomy framework:

- Accuracy: 0.15
- F1 Score: 0.15
- Compliance: 0.20
- Explainability: 0.15
- Edge Readiness: 0.20
- Latency: 0.15

Each weight was independently perturbed by ±0.03, corresponding to the maximum deviation observed during leave-one-out sensitivity testing. Following each perturbation, the remaining weights were proportionally adjusted to preserve the normalisation constraint such that the total weight sum remained equal to 1.

For every perturbed weighting configuration:

- The Clinical Deployment Index (CDI) for HIMS-CDI_cal was recomputed.
- The maximum absolute deviation from the baseline CDI was recorded.
- Governance-gate outcomes were evaluated to determine whether any threshold condition changed status.

Results

- **Baseline CDI for HIMS-CDI_cal:** 0.77 (reported in Table 5 of the main manuscript).
- Across all single-dimension perturbations of ±0.03, the CDI remained within the interval **[0.754, 0.788]**.
- The provisional CDI governance threshold (**CDI ≥ 0.70**) remained satisfied across all perturbation configurations.
- No perturbation caused:
    - violation of edge-feasibility constraints,
    - failure of latency or memory thresholds, or

    - reduction of the Clause Coverage Rate below 0.80.
- All three governance gates, therefore, remained satisfied throughout the complete perturbation analysis.

Interpretation

The limited variability observed across all perturbation scenarios demonstrates that the governance framework is not excessively sensitive to moderate weighting uncertainty. Importantly, the observed perturbations exceeded the magnitude of disagreement identified during the Delphi elicitation phase, thereby representing a conservative robustness test.

The persistence of all governance-gate outcomes further indicates that deployment recommendations are driven primarily by stable model characteristics rather than by fragile weighting assumptions.

Conclusion

The governance-gate outcomes remain robust under AHP weight perturbations substantially larger than the degree of expert disagreement observed during Delphi elicitation. Consequently, the subjective nature of the AHP weighting process does not materially affect the principal finding that the calibrated HIMS-CDI model satisfies the specification-derived deployment criteria.